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WEEK-1: HTML Frames: Frame Structure: Design a webpage for the online book store using HTML frames. Include a navigation frame, a content frame for displaying book details, and a footer frame for additional information.

WEEK1.html:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Online Book Store</title>
  </head>
  <frameset rows="20%,70%,10%" border="2">
    <iframe src=""></iframe>
    <frame src="navigation.html" name="navFrame">
    <frame src="content.html" name="contentFrame">
    <frame src="footer.html" name="footerFrame">
  </frameset>
</html>
```

Navigation.html:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Navigation</title>
  </head>
  <body bgcolor="BBBBBB">
    <h3>Book Categories</h3>
    <h1>Hello World Welcome to VNRVJIET</h1>
    <ul>
      <li><a href="https://www.microsoft.com" target="contentFrame">Computer Science</a></li>
      <li><a href="https://www.ieee.org" target="contentFrame">Electronics</a></li>
      <li><a href="content.html" target="contentFrame">Mechanical</a></li>
      <li><a href="content.html" target="contentFrame">Civil</a></li>
    </ul>
  </body>
</html>
```

Footer.html:

```
<!DOCTYPE html>
<head>
  <title>Footer</title>
</head>
<body bgcolor="90EE90" align="center">
  <p>2025 Online Book Store | Contact: Support@bookstore.com</p>
</body>
</html>
```

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Content.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Book Store</title>
  <link rel="stylesheet" href="content.css">
</head>
<body>

  <section class="categories">
    <h2>Book Categories</h2>
    <ul>
      <li>Fiction</li>
      <li>Non-Fiction</li>
      <li>Science</li>
      <li>Technology</li>
      <li>Children</li>
    </ul>
  </section>

  <section class="books">
    <h2>Available Books</h2>
    <table>
      <tr>
        <th>Book Name</th>
        <th>Author</th>
        <th>Price</th>
        <th>Buy</th>
      </tr>
      <tr>
        <td>The Alchemist</td>
        <td>Paulo Coelho</td>
        <td>$12</td>
        <td><button>Buy</button></td>
      </tr>
      <tr>
        <td>Atomic Habits</td>
        <td>James Clear</td>
        <td>$15</td>
        <td><button>Buy</button></td>
      </tr>
      <tr>
        <td>Clean Code</td>
        <td>Robert C. Martin</td>
        <td>$20</td>
        <td><button>Buy</button></td>
      </tr>
    </table>
  </section>
</body>
```

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```
</table>
</section>

<section class="order">
  <h2>Order a Book</h2>
  <form>
    <label>Full Name</label>
    <input type="text" required>

    <label>Email</label>
    <input type="email" required>

    <label>Select Book</label>
    <select>
      <option>The Alchemist</option>
      <option>Atomic Habits</option>
      <option>Clean Code</option>
    </select>

    <label>Quantity</label>
    <input type="number" min="1" value="1">

    <button type="submit">Place Order</button>
  </form>
</section>

</body>
</html>
```

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The screenshot displays a web browser window with the address bar showing `http://127.0.0.1:5500/week1/week1.html`. The page content is divided into two main sections. The top section, titled "2025 Online Book Store", features a header with "Book Categories" and a "Hello World Welcome to VNRVJIET" message. Below this, there are links for "Computer Science" and "Electronics". A "Book Categories" list includes Fiction, Non-Fiction, Science, Technology, and Children. An "Available Books" table lists three books: "The Alchemist" by Paulo Coelho (\$12), "Atomic Habits" by James Clear (\$15), and "Clean Code" by Robert C. Martin (\$20). Each book has a "Buy" button. Below the table is an "Order a Book" form with fields for "Full Name", "Email", a "Select Book" dropdown (currently showing "The Alchemist"), a "Quantity" dropdown (set to 1), and a "Place Order" button. The bottom section of the browser window shows a Microsoft advertisement for "Copilot+PC" featuring a Surface Pro. The ad includes the text "Meet Surface Pro" and "This laptop's unrivalled flexibility and AI features like Live Captions and Cocreator, enable you to do more than you ever imagined." with a "Learn more" button. The Microsoft logo and navigation links (Microsoft 365, Azure, Copilot, Windows, Surface, Xbox, Support) are visible at the top of the ad. The footer of the browser window displays "2025 Online Book Store | Contact: Support@bookstore.com".

Book Categories

Hello World Welcome to VNRVJIET

- Computer Science
- Electronics

Book Categories

- Fiction
- Non-Fiction
- Science
- Technology
- Children

Available Books

Book Name	Author	Price	Buy
The Alchemist	Paulo Coelho	\$12	Buy
Atomic Habits	James Clear	\$15	Buy
Clean Code	Robert C. Martin	\$20	Buy

Order a Book

Full Name Email Select Book The Alchemist Quantity 1

2025 Online Book Store | Contact: Support@bookstore.com

Book Categories

Hello World Welcome to VNRVJIET

- Computer Science
- Electronics

msn + Bing Breaking news from around the world
Get the Bing + MSN extension No thanks Add it now

Microsoft Microsoft 365 Azure Copilot Windows Surface Xbox Support All Microsoft Search Cart Sign in

Meet Surface Pro

This laptop's unrivalled flexibility and AI features like Live Captions and Cocreator, enable you to do more than you ever imagined.

[Learn more](#)

Copilot+PC

2025 Online Book Store | Contact: Support@bookstore.com

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WEEK-2: CSS Styling: Apply CSS styling to enhance the visual appeal of the frames and content. Use appropriate styles for fonts, colors, and layout.

Week2.html:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Online Book Store</title>
  </head>
  <frameset rows="20%,70%,10%" border="2">
    <iframe src=""></iframe>
    <frame src="navigation.html" name="navFrame">
    <frame src="content.html" name="contentFrame">
    <frame src="footer.html" name="footerFrame">
  </frameset>
</html>
```

Navigation.html:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Navigation</title>
  </head>
  <body bgcolor="BBBBBB">
    <h3>Book Categories</h3>
    <h1>Hello World Welcome to VNRVJIET</h1>
    <ul>
      <li><a href="https://www.google.com" target="contentFrame">Computer Science</a></li>
      <li><a href="content.html" target="contentFrame">Electronics</a></li>
      <li><a href="content.html" target="contentFrame">Mechanical</a></li>
      <li><a href="content.html" target="contentFrame">Civil</a></li>
    </ul>
  </body>
</html>
```

Navigation.css:

```
body {
  font-family: Arial, sans-serif;
  background-color: #eeeeee;
  padding: 10px;
}

h3 {
  color: #2c3e50;
}

ul {
  padding-left: 15px;
}
```

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```
}
```

```
li {  
    margin: 6px 0;  
}
```

```
a {  
    text-decoration: none;  
    color: #2980b9;  
}
```

```
a:hover {  
    text-decoration: underline;  
}
```

Footer.html:

```
<!DOCTYPE html>  
<head>  
    <title>Footer</title>  
</head>  
<body bgcolor="#90EE90" align="center">  
    <p>2025 Online Book Store | Contact: Support@bookstore.com</p>  
</body>  
</html>
```

Footer.css:

```
body {  
    font-family: Arial, sans-serif;  
    background-color: #90EE90;  
    margin: 0;  
    text-align: center;  
}
```

```
p {  
    font-size: 14px;  
    padding-top: 10px;  
}
```

Content.html:

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
    <meta charset="UTF-8">  
    <title>Book Store</title>  
    <link rel="stylesheet" href="content.css">  
</head>  
<body>  
  
    <!-- Categories List -->
```

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```
<section class="categories">
  <h2>Book Categories</h2>
  <ul>
    <li>Fiction</li>
    <li>Non-Fiction</li>
    <li>Science</li>
    <li>Technology</li>
    <li>Children</li>
  </ul>
</section>

<!-- Books Table -->
<section class="books">
  <h2>Available Books</h2>
  <table>
    <tr>
      <th>Book Name</th>
      <th>Author</th>
      <th>Price</th>
      <th>Buy</th>
    </tr>
    <tr>
      <td>The Alchemist</td>
      <td>Paulo Coelho</td>
      <td>$12</td>
      <td><button>Buy</button></td>
    </tr>
    <tr>
      <td>Atomic Habits</td>
      <td>James Clear</td>
      <td>$15</td>
      <td><button>Buy</button></td>
    </tr>
    <tr>
      <td>Clean Code</td>
      <td>Robert C. Martin</td>
      <td>$20</td>
      <td><button>Buy</button></td>
    </tr>
  </table>
</section>

<!-- Order Form -->
<section class="order">
  <h2>Order a Book</h2>
  <form>
    <label>Full Name</label>
    <input type="text" required>
```

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```
<label>Email</label>
<input type="email" required>

<label>Select Book</label>
<select>
  <option>The Alchemist</option>
  <option>Atomic Habits</option>
  <option>Clean Code</option>
</select>

<label>Quantity</label>
<input type="number" min="1" value="1">

<button type="submit">Place Order</button>
</form>
</section>

</body>
</html>
```

Content.css:

```
/* General page styling */
body {
  font-family: Arial, sans-serif;
  background-color: #f5f7fa;
  margin: 0;
  padding: 15px;
  color: #333;
}

/* Section styling */
section {
  background: #ffffff;
  padding: 15px;
  margin-bottom: 20px;
  border-radius: 6px;
  box-shadow: 0 2px 6px rgba(0,0,0,0.1);
}

/* Headings */
h2 {
  color: #2c3e50;
  margin-bottom: 10px;
  border-bottom: 2px solid #3498db;
  padding-bottom: 5px;
}

/* Categories list */
.categories ul {
```


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```
list-style-type: square;
padding-left: 20px;
}

.categories li {
  margin: 5px 0;
}
/* Table styling */
table {
  width: 100%;
  border-collapse: collapse;
}

th {
  background-color: #3498db;
  color: white;
  padding: 8px;
}

td {
  padding: 8px;
  border-bottom: 1px solid #ddd;
}

/* Buttons */
button {
  background-color: #2ecc71;
  color: white;
  border: none;
  padding: 6px 12px;
  border-radius: 4px;
  cursor: pointer;
}

button:hover {
  background-color: #27ae60;
}

/* Form styling */
form {
  display: grid;
  row-gap: 8px;
}

input, select {
  padding: 6px;
  border: 1px solid #ccc;
  border-radius: 4px;
}
```

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WEBPAGE:

Book Categories

Hello World Welcome to VNRVJIET

- Computer Science
- Electronics

Book Categories

- Fiction
- Non-Fiction
- Science
- Technology
- Children

Available Books

Book Name	Author	Price	Buy
The Alchemist	Paulo Coelho	\$12	Buy
Atomic Habits	James Clear	\$15	Buy
Clean Code	Robert C. Martin	\$20	Buy

2025 Online Book Store | Contact: Support@bookstore.com

Book Categories

Hello World Welcome to VNRVJIET

- Computer Science
- Electronics

The Alchemist	Paulo Coelho	\$12	Buy
Atomic Habits	James Clear	\$15	Buy
Clean Code	Robert C. Martin	\$20	Buy

Order a Book

Full Name

Email

Select Book

The Alchemist

Quantity

Place Order

2025 Online Book Store | Contact: Support@bookstore.com

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WEEK-3:

JavaScript:

a) User Registration Form

Create a registration form for users signing up on the online bookstore. Validate the following:

- **Username:** Should not be empty and must be at least 5 characters long.
- **Password:** Should not be empty and must be at least 8 characters long.
- **Confirm Password:** Should match the entered password.

b) Forgot Password Form

Implement a form for users to reset their password. Validate the following:

- **Email Address:** Should be a valid email format.
- **Provide feedback** if the email address is not associated with any account.

c) Dynamically Content Loading

Use JavaScript to dynamically load book details into the content frame when a user clicks on a book in the navigation frame.

Week3.html:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Online Book Store - Week 3</title>
  </head>
  <frameset rows="20%,70%,10%" border="2">
    <frame src="navigation.html" name="navFrame">
    <frame src="content.html" name="contentFrame">
    <frame src="footer.html" name="footerFrame">
  </frameset>
</html>
```

Week3.js:

// week3.js - Dynamically load book details into the content frame

```
document.addEventListener('DOMContentLoaded', function () {
  // This script runs inside the navigation frame (navigation.html)
  var navLinks = document.querySelectorAll('a[data-book-id]');
  if (!navLinks.length) {
    return;
  }

  var bookData = {
    'clean-code': {
      title: 'Clean Code',
      author: 'Robert C. Martin',
      category: 'Technology / Programming',
      price: '$20',
      isbn: '978-0132350884',
    }
  }
```

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```
        availability: 'In Stock',
        description: 'A handbook of agile software craftsmanship focusing on writing clean,
maintainable code.'
```

```
    },
```

```
    'alchemist': {
```

```
        title: 'The Alchemist',
```

```
        author: 'Paulo Coelho',
```

```
        category: 'Fiction',
```

```
        price: '$12',
```

```
        isbn: '978-0061122415',
```

```
        availability: 'In Stock',
```

```
        description: 'A novel about a shepherd boy on a journey to realize his personal legend.'
```

```
    },
```

```
    'atomic-habits': {
```

```
        title: 'Atomic Habits',
```

```
        author: 'James Clear',
```

```
        category: 'Self Help',
```

```
        price: '$15',
```

```
        isbn: '978-0735211292',
```

```
        availability: 'Limited Stock',
```

```
        description: 'Practical strategies for building good habits and breaking bad ones.'
```

```
    },
```

```
    'ds-algo': {
```

```
        title: 'Data Structures and Algorithms',
```

```
        author: 'Narasimha Karumanchi',
```

```
        category: 'Computer Science',
```

```
        price: '$25',
```

```
        isbn: '978-8193245279',
```

```
        availability: 'Out of Stock',
```

```
        description: 'Detailed explanations of core data structures and algorithms with interview
focus.'
```

```
    }
```

```
};
```

```
function loadBookIntoContentFrame(bookId) {
```

```
    var info = bookData[bookId];
```

```
    if (!info) {
```

```
        return;
```

```
    }
```

```
    var contentFrame = parent.frames['contentFrame'];
```

```
    if (!contentFrame || !contentFrame.document) {
```

```
        return;
```

```
    }
```

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```
var doc = contentFrame.document;
var container = doc.getElementById('bookDetails');
if (!container) {
    return;
}

container.innerHTML =
    '<h3>' + info.title + '</h3>' +
    '<p><strong>Author:</strong> ' + info.author + '</p>' +
    '<p><strong>Category:</strong> ' + info.category + '</p>' +
    '<p><strong>ISBN:</strong> ' + info.isbn + '</p>' +
    '<p><strong>Price:</strong> ' + info.price + '</p>' +
    '<p><strong>Availability:</strong> ' + info.availability + '</p>' +
    '<p><strong>Description:</strong> ' + info.description + '</p>';
}

navLinks.forEach(function (link) {
    link.addEventListener('click', function (e) {
        e.preventDefault();
        var bookId = this.getAttribute('data-book-id');
        loadBookIntoContentFrame(bookId);
    });
});
```

Registration.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>User Registration - Online Bookstore</title>
    <link rel="stylesheet" href="registration.css">
</head>
<body>
    <div class="container">
        <h2>User Registration</h2>
        <form id="registrationForm" autocomplete="off">
            <label for="username">Username:</label>
            <input type="text" id="username" name="username" required>

            <label for="password">Password:</label>
            <input type="password" id="password" name="password" required>

            <label for="confirmPassword">Confirm Password:</label>
            <input type="password" id="confirmPassword" name="confirmPassword" required>
        </form>
    </div>
</body>
</html>
```

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```
<div id="errorMsg" class="error" style="display:none;"></div>
<button type="submit">Register</button>
<div style="text-align:center; margin-top:15px;">
  <a href="forgotpassword.html">Forgot Password?</a>
</div>
</form>
</div>
<script src="registration.js"></script>
</body>
</html>
```

ForgotPassword.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Forgot Password - Online Bookstore</title>
  <link rel="stylesheet" href="forgotpassword.css">
</head>
<body>
  <div class="container">
    <h2>Forgot Password</h2>
    <form id="forgotPasswordForm" autocomplete="off">
      <label for="email">Email Address:</label>
      <input type="email" id="email" name="email" required>
      <div id="fpErrorMsg" class="error" style="display:none;"></div>
      <button type="submit">Reset Password</button>
    </form>
  </div>
  <script src="forgotpassword.js"></script>
</body>
</html>
```

Forgotpassword.js:

```
document.addEventListener('DOMContentLoaded', function() {
  var form = document.getElementById('forgotPasswordForm');
  if (!form) return;
  form.addEventListener('submit', function(e) {
    e.preventDefault();
    var email = document.getElementById('email').value.trim();
    var errorMsg = document.getElementById('fpErrorMsg');
    errorMsg.style.display = 'none';
    errorMsg.innerText = "";
  });
});
```

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```
var emailRegex = /^[w-\.]++@([\w-]+\.)+[\w-]{2,4}$/;
if (!emailRegex.test(email)) {
    errorMsg.innerText = 'Please enter a valid email address.';
    errorMsg.style.display = 'block';
    return;
}

// Simulated check for email association (for demo, only these emails are valid)
var validEmails = ['user1@example.com', 'user2@example.com',
'test@bookstore.com', 'pranav.tungani@gmail.com'];
if (!validEmails.includes(email)) {
    errorMsg.innerText = 'This email address is not associated with any account.';
    errorMsg.style.display = 'block';
    return;
}

alert('A password reset link has been sent to your email.');
```

```
form.reset();
});
});
```

Registration.js:*// registration.js - Security and validation for registration form*

```
document.addEventListener('DOMContentLoaded', function() {
    var form = document.getElementById('registrationForm');
    if (!form) return;
    form.addEventListener('submit', function(e) {
        e.preventDefault();
        var username = document.getElementById('username').value.trim();
        var password = document.getElementById('password').value;
        var confirmPassword = document.getElementById('confirmPassword').value;
        var errorMsg = document.getElementById('errorMsg');
        errorMsg.style.display = 'none';
        errorMsg.innerText = "";

        // Username validation
        if (username.length < 5) {
            errorMsg.innerText = 'Username must be at least 5 characters long.';
            errorMsg.style.display = 'block';
            return;
        }

        // Password validation
        if (password.length < 8) {
            errorMsg.innerText = 'Password must be at least 8 characters long.';
            errorMsg.style.display = 'block';
        }
    });
});
```

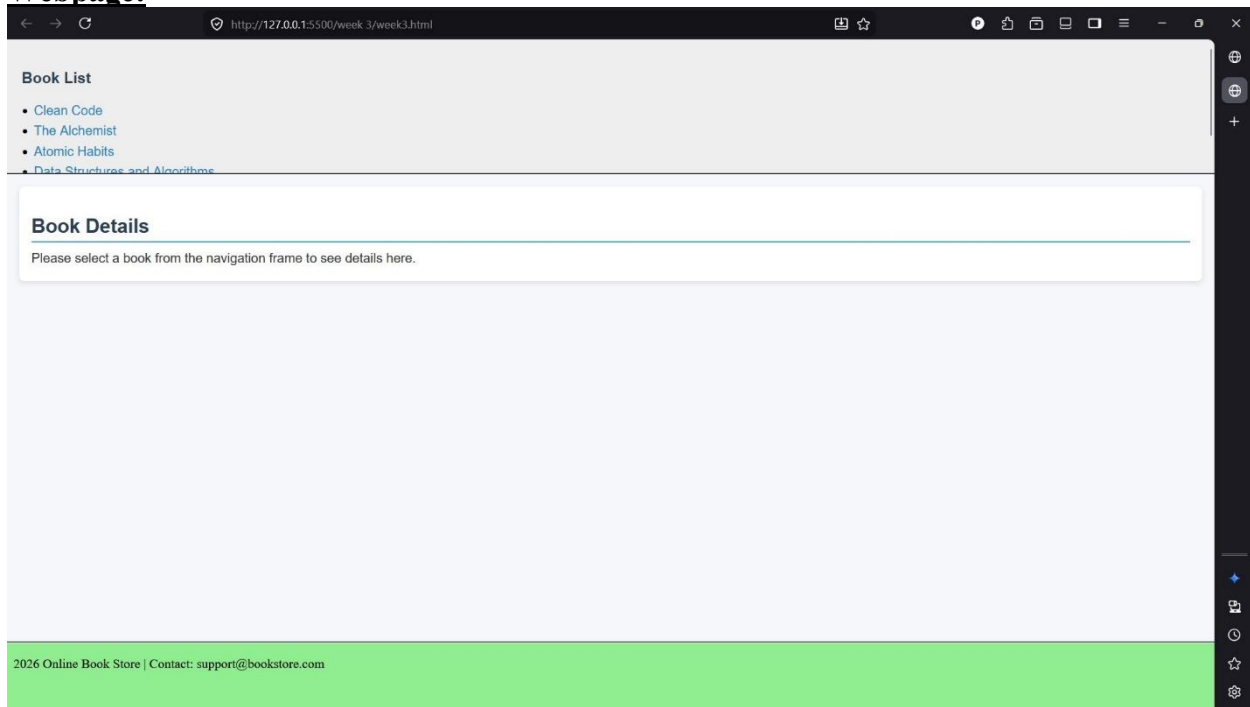
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```
        return;
    }
    // Password strength (at least 1 uppercase, 1 lowercase, 1 number)
    var strongRegex = /^(?=.*[a-z])(?=.*[A-Z])(?=.*\d).+$/;
    if (!strongRegex.test(password)) {
        errorMsg.innerText = 'Password must contain uppercase, lowercase, and a number.';
        errorMsg.style.display = 'block';
        return;
    }
    // Confirm password
    if (password !== confirmPassword) {
        errorMsg.innerText = 'Passwords do not match.';
        errorMsg.style.display = 'block';
        return;
    }
    // All validations passed
    alert('Registration successful!');
    form.reset();

    // Navigate to the content page after successful registration
    window.location.href = 'navigation.html';
});
});
```

Webpage:

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The image displays two browser windows side-by-side, showing a web application interface. The top window shows a 'Book List' and 'Book Details' for 'Atomic Habits'. The bottom window shows a 'User Registration' form with fields for Username, Password, and Confirm Password, along with a 'Register' button and a 'Forgot Password?' link. Below this, another browser window shows a 'Forgot Password' form with an 'Email Address' field and a 'Reset Password' button.

Book List

- Clean Code
- The Algorithm
- Atomic Habits
- Data Structures and Algorithms

Book Details

Atomic Habits

Author: James Clear
Category: Self-Help
ISBN: 978-0735213292
Price: \$15
Availability: In Stock
Description: Practical strategies for building good habits and breaking bad ones.

User Registration

Username:
user1@gmail.com

Password:

Confirm Password:

[Register](#)

[Forgot Password?](#)

Forgot Password

Email Address:

[Reset Password](#)

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WEEK-4: Shopping Cart Functionality: Implement a simple shopping cart using JavaScript. Allow users to add books to their cart and view the cart's contents.

Cart.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>My Cart</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <main>
    <h1>My Cart</h1>
    <a href="index.html">Back to Store</a>

    <div id="cart-items"></div>
    <p><strong>Total: Rs. <span id="total">0</span></strong></p>
    <button onclick="clearCart()">Clear Cart</button>
  </main>

  <script src="script.js"></script>
</body>
</html>
```

Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Book Store</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <main>
    <h1>Book Store</h1>
    <a href="cart.html">View Cart (<span id="cart-count">0</span>)</a>

    <div id="books">
      <div class="book">
        <h3>The Alchemist</h3>
        <p>By Paulo Coelho</p>
        <p>Price: Rs. 299</p>
        <button onclick="addToCart('The Alchemist', 299)">Add to Cart</button>
      </div>
    </div>
  </main>
</body>
</html>
```

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```
<div class="book">
  <h3>Atomic Habits</h3>

  <p>By James Clear</p>
  <p>Price: Rs. 499</p>
  <button onclick="addToCart('Atomic Habits', 499)">Add to Cart</button>
</div>

<div class="book">
  <h3>Rich Dad Poor Dad</h3>
  <p>By Robert Kiyosaki</p>
  <p>Price: Rs. 349</p>
  <button onclick="addToCart('Rich Dad Poor Dad', 349)">Add to Cart</button>
</div>

<div class="book">
  <h3>Deep Work</h3>
  <p>By Cal Newport</p>
  <p>Price: Rs. 399</p>
  <button onclick="addToCart('Deep Work', 399)">Add to Cart</button>
</div>
</div>
</main>

<script src="script.js"></script>
</body>
</html>
```

Script.js:

```
// Get cart from localStorage (or empty array)
function getCart() {
  return JSON.parse(localStorage.getItem('cart')) || [];
}

// Save cart to localStorage
function saveCart(cart) {
  localStorage.setItem('cart', JSON.stringify(cart));
}

// Add a book to the cart
function addToCart(title, price) {
  var cart = getCart();
  cart.push({ title: title, price: price });
  saveCart(cart);
  alert(title + ' added to cart!');
  updateCount();
}
```

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```
}
```

```
// Update the cart count shown on the store page
```

```
function updateCount() {  
    var countEl = document.getElementById('cart-count');  
    if (countEl) {  
        countEl.textContent = getCart().length;  
    }  
}
```

```
// Show cart items on cart.html
```

```
function showCart() {  
    var cart = getCart();  
    var container = document.getElementById('cart-items');  
    if (!container) return;  
  
    if (cart.length === 0) {  
        container.innerHTML = '<p>Your cart is empty.</p>';  
        document.getElementById('total').textContent = 0;  
        return;  
    }  
  
    var html = '<ul>';  
    var total = 0;  
    for (var i = 0; i < cart.length; i++) {  
        html += '<li>' + cart[i].title + ' - Rs. ' + cart[i].price + '</li>';  
        total += cart[i].price;  
    }  
    html += '</ul>';  
    container.innerHTML = html;  
    document.getElementById('total').textContent = total;  
}
```

```
// Clear all items from cart
```

```
function clearCart() {  
    localStorage.removeItem('cart');  
    showCart();  
}
```

```
// Run on page load
```

```
updateCount();  
showCart();
```

Styles.css:

```
body {  
    font-family: Arial, sans-serif;
```

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```
margin: 0;
padding: 0;
background: #f5f7fb;
}
/* Books grid */
#books{
  display: flex;
  flex-wrap: wrap;
  gap: 18px;
  justify-content: center;
}

.book {
  background: #ffffff;
  border-radius: 10px;
  box-shadow: 0 2px 8px rgba(11,37,69,0.06);
  padding: 16px;
  width: 240px;
  display: flex;
  flex-direction: column;
  gap: 8px;
  transition: transform 160ms ease, box-shadow 160ms ease;
}

.book:hover {
  transform: translateY(-6px);
  box-shadow: 0 10px 22px rgba(11,37,69,0.10);
}

.book h3 { margin: 0; font-size: 1.05rem; color: #071b2d; }
.book p { margin: 0; color: #475569; font-size: 0.9rem; }
.book .price { font-weight: 700; color: #111827; }

button {
  padding: 10px 14px;
  background-color: #0b74de;
  color: white;
  border: none;
  border-radius: 8px;
  cursor: pointer;
  align-self: start;
  transition: background-color 120ms ease, transform 120ms ease;
}

button:hover { background-color: #075fae; transform: translateY(-2px); }
```

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```
button:active { transform: translateY(0); }
```

```
/* Cart link badge */
```

```
#cart-count {  
  background: #ef4444;  
  color: #fff;  
  padding: 2px 8px;  
  border-radius: 999px;  
  margin-left: 6px;  
  font-weight: 700;  
  font-size: 0.85rem;  
}
```

```
/* Cart page styles */
```

```
#cart-items { margin: 18px 0; }  
.cart-item {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
  padding: 10px 12px;  
  background: #fff;  
  border-radius: 8px;  
  box-shadow: 0 1px 4px rgba(11,37,69,0.04);  
  margin-bottom: 10px;  
}
```

```
.cart-item .name { font-weight: 600; }  
.cart-item .price { color: #0b2545; font-weight: 700; }
```

```
/* Responsive */
```

```
@media (max-width: 600px) {  
  main { margin: 18px; padding: 12px; }  
  .book { width: 100%; max-width: 420px; }  
}
```

```
/* Small utility */
```

```
.muted { color: #6b7280; font-size: 0.9rem; }
```

```
/* Style cart list rendered by script.js */
```

```
#cart-items ul { list-style: none; padding: 0; margin: 0; }  
#cart-items ul li {  
  padding: 10px 12px;  
  background: #fff;  
  border-radius: 8px;  
  box-shadow: 0 1px 4px rgba(11,37,69,0.04);  
}
```

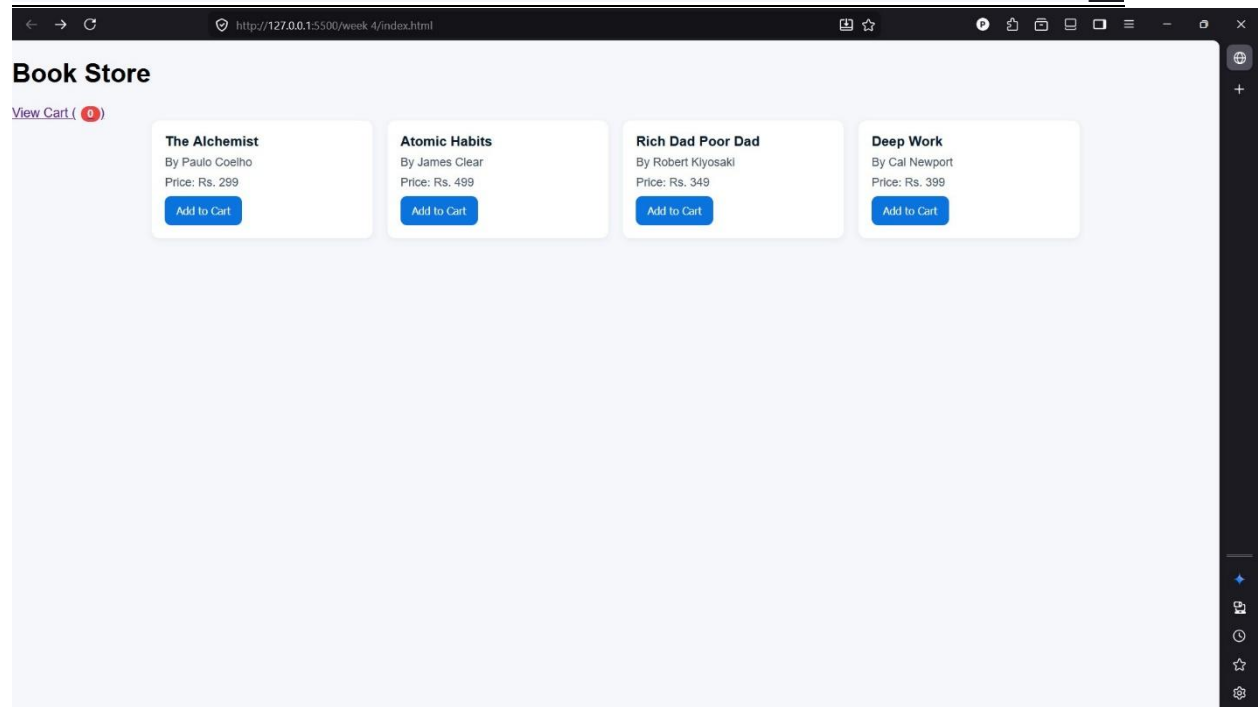
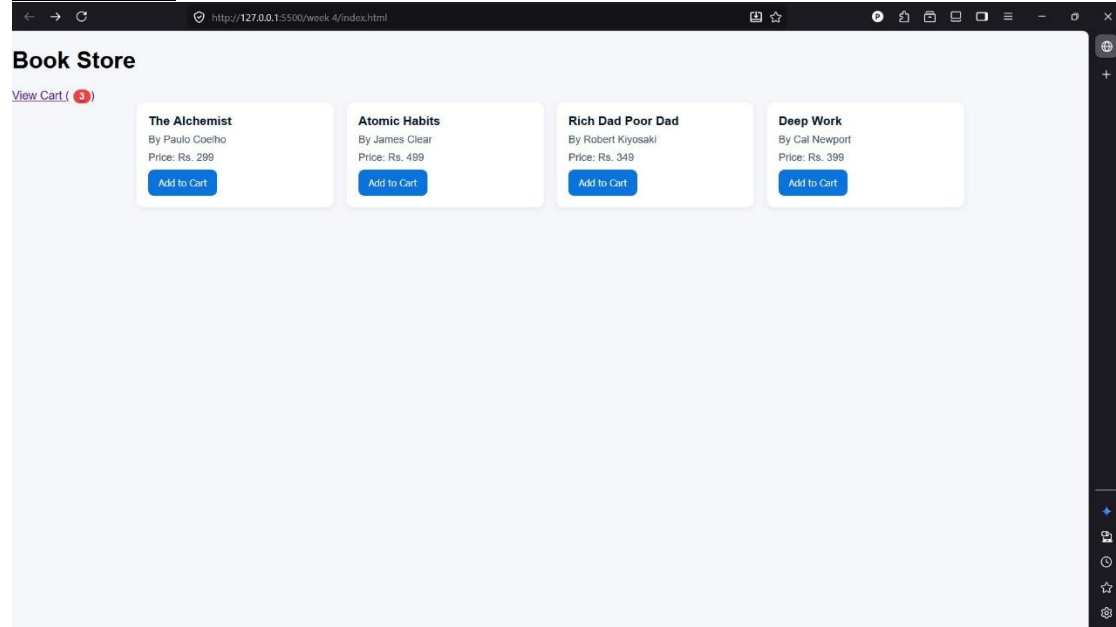
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```
margin-bottom: 10px;
display: flex;
justify-content: space-between;
align-items: center;
}
#cart-items ul li::after { content: attr(data-price); font-weight:700; color:#0b2545; }
```

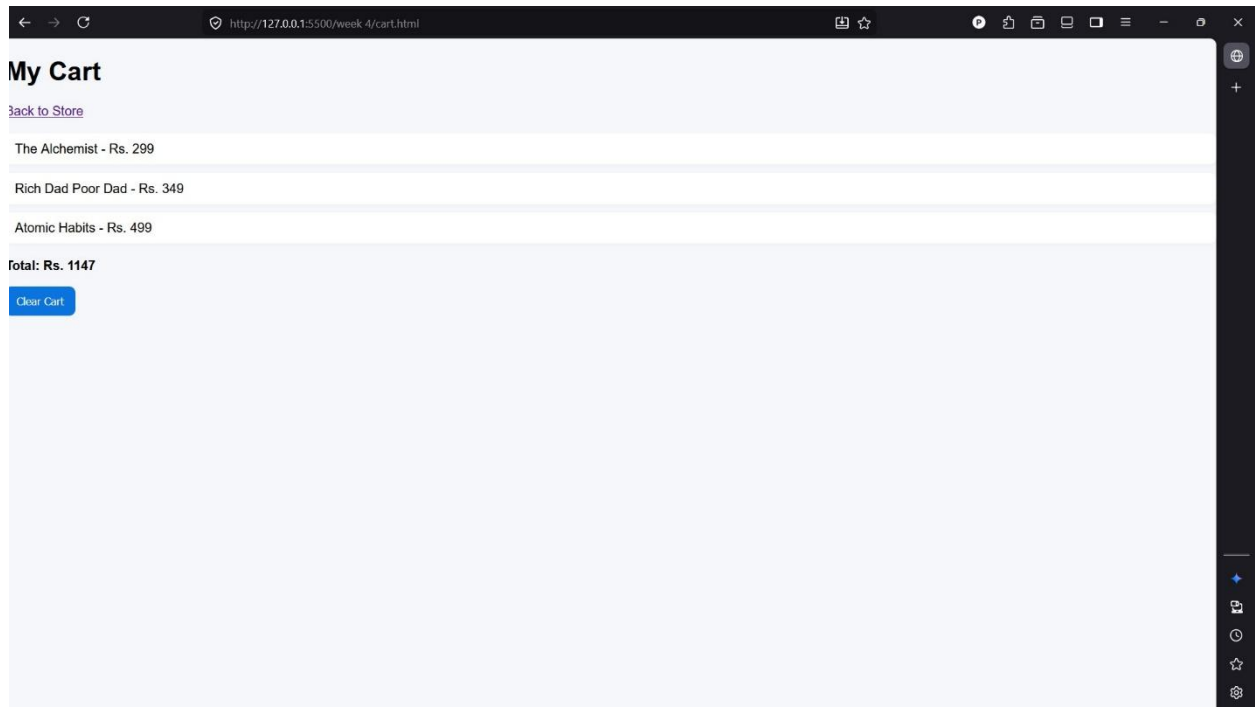
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WEEK-5: XML Structure and Design: Book Catalog XML: Design an XML structure to represent the catalog of books in the online store. Include details such as book title, author, genre, price, and availability.

Book catalog view.html:

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width,initial-scale=1">
  <title>Book Catalog Viewer</title>
  <link rel="stylesheet" href="book_catalog.css">
</head>
<body>
  <header class="site-header">
    <h1>Book Catalog</h1>
  </header>

  <main class="container">
    <div class="controls">
      <input id="search" placeholder="Search by title, author, or genre" />
      <select id="genreFilter"><option value="">All genres</option></select>
      <button id="refresh">Refresh</button>
    </div>

    <div id="catalog" class="grid" aria-live="polite"></div>
  </main>

  <script src="book_catalog.js"></script>
</body>
</html>
```

Book catalog.css:

```
:root {
  --bg:#f6f8fa;
  --card:#ffffff;
  --muted:#6b7280;
  --accent:#2563eb;
  --success:#10b981;
  --danger:#ef4444;
  --glass:rgba(255,255,255,0.6);
}
*{box-sizing:border-box}
body {font-family:Segoe UI,Roboto,Arial,sans-serif;margin:0;background:var(--bg);color:#111}
.site-header {background:linear-gradient(90deg,var(--accent),#7c3aed);color:white;padding:20px 24px}
```

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```
.site-header h1 {margin:0;font-size:1.25rem}
.container {max-width:1100px;margin:24px auto;padding:0 16px}
.controls {display:flex;gap:10px;align-items:center;margin-bottom:18px}
.controls input, .controls select {padding:10px 12px;font-size:0.95rem;border:1px solid
#e6e9ee;border-radius:8px;min-width:0}
.controls button {padding:10px 14px;border-radius:8px;background:var(--
accent);color:white;border:none;cursor:pointer}
.grid {display:grid;grid-template-columns:repeat(auto-fill,minmax(220px,1fr));gap:16px}
.card {background:var(--card);padding:14px;border-radius:12px;box-shadow:0 6px 20px
rgba(20,20,40,0.06);display:flex;flex-direction:column;gap:8px}
.title {font-weight:700;font-size:1.02rem}
.author {color:var(--muted);font-size:0.92rem}
.meta {display:flex;justify-content:space-between;align-items:center;margin-top:6px}
.genre {font-size:0.85rem;padding:6px 8px;border-radius:999px;background:var(--
glass);color:#374151}
.price {font-weight:700;color:var(--accent)}
.availability {font-size:0.85rem;padding:6px 8px;border-radius:999px}
.availability.in-stock {background:rgba(16,185,129,0.08);color:var(--success);border:1px solid
rgba(16,185,129,0.12)}
.availability.out-of-stock {background:rgba(239,68,68,0.08);color:var(--danger);border:1px solid
rgba(239,68,68,0.12)}
.empty {padding:30px;text-align:center;color:var(--muted)}
@media (max-width:520px) { .controls {flex-direction:column;align-items:stretch} .controls
input {width:100%} }
```

Book catalog.js:

```
async function loadCatalog() {
  try {
    const res = await fetch('book_catalog.xml');
    if(!res.ok) throw new Error('Failed to load XML');
    const text = await res.text();
    const parser = new DOMParser();
    const xml = parser.parseFromString(text, 'application/xml');

    const books = Array.from(xml.getElementsByTagName('book')).map(b => {
      const title = b.getElementsByTagName('title')[0]?.textContent || '';
      const author = b.getElementsByTagName('author')[0]?.textContent || '';
      const genre = b.getElementsByTagName('genre')[0]?.textContent || '';
      const priceEl = b.getElementsByTagName('price')[0];
      const price = priceEl?.textContent || '';
      const currency = priceEl?.getAttribute('currency') || '';
      const availEl = b.getElementsByTagName('availability')[0];
      const availabilityCount = availEl?.textContent || '0';
      const status = availEl?.getAttribute('status') || (Number(availabilityCount)>0 ? 'in-stock' : 'out-
of-stock');
```

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```
        return { id: b.getAttribute('id'), title, author, genre, price, currency, availabilityCount:
Number(availabilityCount), status };
    });
```

```
        return books;
    } catch(err){
        console.error(err);
        return null;
    }
}
```

```
function renderBooks(books){
    const container = document.getElementById('catalog');
    container.innerHTML = "";
    if(!books || books.length===0){
        container.innerHTML = '<div class="empty">No books found.</div>';
        return;
    }
}
```

```
const frag = document.createDocumentFragment();
books.forEach(b=>{
    const card = document.createElement('article');
    card.className = 'card';
    card.innerHTML = `
        <div class="title">${escapeHtml(b.title)}</div>
        <div class="author">by ${escapeHtml(b.author)}</div>
        <div class="meta">
            <div class="genre">${escapeHtml(b.genre)}</div>
            <div class="price">${escapeHtml(b.currency)} ${escapeHtml(b.price)}</div>
        </div>
        <div style="display:flex;justify-content:space-between;align-items:center">
            <div class="availability ${b.status === 'in-stock' ? 'in-stock' : 'out-of-stock'}">${b.status ===
'in-stock' ? 'In stock' : 'Out of stock'}</div>
            <div style="color:var(--muted);font-size:0.85rem">Qty: ${b.availabilityCount}</div>
        </div>
    `;
    frag.appendChild(card);
});
container.appendChild(frag);
}
```

```
function escapeHtml(s){
    return String(s).replaceAll('&', '&amp;').replaceAll('<', '&lt;').replaceAll('>', '&gt;');
}
```

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```
function applyFilters(allBooks){
  const q = document.getElementById('search').value.trim().toLowerCase();
  const genre = document.getElementById('genreFilter').value;
  let filtered = allBooks.filter(b => {
    const hay = `${b.title} ${b.author} ${b.genre}`.toLowerCase();
    return hay.includes(q) && (genre === "" || b.genre === genre);
  });
  renderBooks(filtered);
}

(async function init(){
  const allBooks = await loadCatalog();
  if(allBooks === null){
    document.getElementById('catalog').innerHTML = '<div class="empty">Could not load
catalog. Make sure <code>book_catalog.xml</code> is next to the viewer.</div>';
    return;
  }

  // populate genre filter
  const genres = Array.from(new Set(allBooks.map(b=>b.genre).filter(Boolean))).sort();
  const select = document.getElementById('genreFilter');
  genres.forEach(g=>{
    const opt = document.createElement('option');
    opt.value = g; opt.textContent = g; select.appendChild(opt);
  });

  // initial render
  renderBooks(allBooks);

  // wire controls
  document.getElementById('search').addEventListener('input', ()=>applyFilters(allBooks));
  select.addEventListener('change', ()=>applyFilters(allBooks));
  document.getElementById('refresh').addEventListener('click', async ()=>{
    const refreshed = await loadCatalog();
    if(refreshed) {
      allBooks.length = 0; allBooks.push(...refreshed);
      applyFilters(allBooks);
    }
  });
})();
```

Book catalog.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<catalog>
  <book id="B001">
```

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```
<title>The Great Gatsby</title>
<author>F. Scott Fitzgerald</author>
<genre>Fiction</genre>
<price currency="USD">10.99</price>
<availability status="in-stock">15</availability>
</book>
```

```
<book id="B002">
  <title>A Brief History of Time</title>
  <author>Stephen Hawking</author>
  <genre>Science</genre>
  <price currency="USD">15.50</price>
  <availability status="in-stock">8</availability>
</book>
```

```
<book id="B003">
  <title>Clean Code</title>
  <author>Robert C. Martin</author>
  <genre>Technology</genre>
  <price currency="USD">32.00</price>
  <availability status="in-stock">23</availability>
</book>
```

```
<book id="B004">
  <title>To Kill a Mockingbird</title>
  <author>Harper Lee</author>
  <genre>Fiction</genre>
  <price currency="USD">12.99</price>
  <availability status="out-of-stock">0</availability>
</book>
```

```
<book id="B005">
  <title>Sapiens: A Brief History of Humankind</title>
  <author>Yuval Noah Harari</author>
  <genre>History</genre>
  <price currency="USD">18.75</price>
  <availability status="in-stock">5</availability>
</book>
</catalog>
```

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